AGENTX HACKATHON 2026 · TRACK PS-04

ReturnLens

“Every return decision, backed by evidence.”

Connect customer evidence, order facts and policy — then let the seller decide.

ReturnLens unifies unboxing video evidence, store truth, and policy checks into a 1-click decision console for solo merchants.

Team Rookie · AgentX PS-04 · 24h Prototype

RETURNLENS SELLER CONSOLE — CASE #RT-8821



ORDERED: HEAD-WHT-02 (WHITE)



OBSERVED: BLACK ITEM · FRAME 00:07

P-03: COLOR CONTRADICTION DETECTED
P-04 Return Window: PASS (Day 4 of 7) | P-03 Manifest: CONTRADICTION FLAGGED

SELLER REVIEW REQUIRED (HUMAN GATE)

Review Evidence & Authorize Draft

THE PROBLEM · SOLO SELLER BOTTLENECK

One return. Four disconnected sources.

Four systems. No traceable decision. Without a shared record, the seller connects order facts, video, and policy manually from memory.

01 BUYER MESSAGE
Subjective text & unindexed video claim in chat

02 ORDER RECORD
Shopify manifest: HEAD-WHT-02, Matte White, ₹1,499

03 COURIER STATUS
Shippo scan: delivered 4 days ago (Day 4 of 7)

04 RETURN POLICY
7-day defect window; security packaging tag rule

THE MISSING LINK

NO SHARED EVIDENCE RECORD

"What evidence supports this return decision?"

Four disconnected sources. One seller making the call.

The gap is the absence of an auditable artifact linking order truth, video keyframes, and policy before money moves.

Fraud research: 9% strict fraud (NRF 2025) vs 15.14% fraud + abuse (Appriss/Deloitte 2024). Details in notes.

STAR DEMO · CORE BENCHMARK SCENARIO

The order says white. The video shows black.

ReturnLens flags the contradiction — the seller decides.



STORE CATALOG TRUTH: HEAD-WHT-02 (WHITE)



UNBOXING KEYFRAME (STAGED): MIDNIGHT BLACK

Visual Grounding: Keyframe 00:07 confirms black casing. Exact SKU barcode is unverified in this frame.

Frame citation provides a traceable timestamped reference, not absolute proof of video authenticity.

POLICY GATE EVALUATION

COLOR CONTRADICTION

- ORDERED — Matte White
- OBSERVED — Midnight Black
- EXACT SKU — Not verified

SELLER REVIEW REQUIRED

No automatic denial or refund without seller authorization.

Financial safety ceiling:
`0 < amount ≤ min(₹4,999, remaining, paid)`

SYSTEM ARCHITECTURE · DYNAMIC AGENTIC GOVERNANCE

Models observe. Code verifies. Seller authorizes.

Dynamic action selection and stateful resumption under deterministic human authorization.

MODEL

Plan and observe

Planner proposes typed actions · Vision model generates cited observations.

CODE

Retrieve, validate and enforce rules

Backend code validates schemas · FFmpeg extracts frames · Python code evaluates rules P-01 to P-08.

SELLER

Approve consequential actions

Fail-closed authorization gate · 1-click human merchant confirmation before refund release.

DYNAMIC STATEFUL CLARIFICATION LOOP (CASE C)

DARK EVIDENCE → ASK FOR PHOTO → WAITING_FOR_EVIDENCE → CUSTOMER UPLOAD → RESUME CASE → SELLER REVIEW

Agent calls ask_for_evidence() on low-light frames, serializes context, and resumes execution upon customer upload.

SECONDARY PS-04 ROUTE ("Where is my order?"): Store API → Courier API → Customer Reply (Zero vision overhead)

Structured JSON Audit Log recorded at every state transition.

DEMO SCENARIOS · BEHAVIORAL MATRIX

Different evidence. Different action.

Predictable, controlled prototype responses across four distinct evidence conditions.

01 · CONSISTENT	**Evidence Supported · Manifest matches, packaging tag present, cosmetic defect observed**	→	**PREPARE SELLER REVIEW** Drafts mock return label and refund draft in pending_return_receipt. Money releases only on physical receipt.
02 · CONTRADICTION	**Ordered White, Observed Black · HEAD-WHT-02 ordered vs Frame 00:07 Black item**	→	**ESCALATE TO SELLER** Policy P-03 blocks automation. Inhibits label and refund. Seller selects: mis-shipment / box label / warehouse inspection.
03 · UNCLEAR	**Insufficient Visual Evidence · 4s low-light video (brightness 28/255), barcode obscured**	→	**ASK AND RESUME** Calls ask_for_evidence(); transitions to WAITING_FOR_EVIDENCE; resumes context on upload (1 retry limit).
04 · SEQUENCE	**Continuity Uncertain · Staged take from sealed parcel opening to product reveal**	→	**INSPECT SEQUENCE CONTINUITY** Expected prototype behavior: inspect sequence continuity and flag cutaways for seller review.

A credible prototype in 24 hours.

Strict scope boundaries and single-threaded ownership make the prototype buildable in one day.

00–06h

Mock APIs & Agent State

Person A (Agent / Backend / Policy)

LangGraph state machine, mock store & courier fixtures, order status fast path.

06–12h

Evidence Pipeline & Policy

Person B (Vision / Evidence / UI)

FFmpeg keyframe indexer, vision prompt schema, critic prompt, seller console UI.

12–18h

State Flow & Scenarios

Person C (Fixtures / Integration / QA)

Clarification/resume loop, wiring 4 staged demo cases, end-to-end event bus.

18–24h

Testing & Submission

All 3 Members

Authorization safety tests, edge cases, fallback validation, demo video recording.

MUST DELIVER (CORE PROTOTYPE)

- Mock store & courier APIs
- Deterministic policy & seller gate
- Color-mismatch contradiction scenario
- Clarification / resume scenario (Case C)
- Structured JSON audit trail log
- Order-status fast path lookup

CUT FIRST IF BEHIND (DEFERRED EXPANSION)

- Advanced continuity checks (Case D stretch)
- Live external carrier & store integrations
- Video OCR barcode scanner
- Merchant analytics dashboard